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Modified dispersion relations from gravitational wave signal detections in multiple frequency bands

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ABSTRACT
Recent direct detection of gravitational waves has made possible a new era of astronomy; in this paper, we establish the current state of the science and discuss how gravitational waves produced by different classes of objects can inform our understanding of fundamental physics. We outline also the project goal of developing a phenomenological model for modified dispersion relations resulting from measurements of gravitational waves in multiple frequency bands.

1. Introduction

Gravitational waves quantify the energy released by an accelerating object as predicted by general relativity (1). In the case of a binary system, as the component masses approach each other, they distort spacetime radially in every direction, and this distortion can be observed as waves propagating at the speed of light (2). The inspiral of these component masses m_1 and m_2, and their subsequent merger results in a coalesced object described by a chirp mass,

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} = \frac{c^3}{G}\left[\frac{5}{96}\pi^{-8/3} f^{-11/3}\dot{f}\right]^{3/5} \tag{1}$$

where f is its observed frequency, and G and c the familiar constants (1). This chirp mass is effectively the total mass of the system once energy loss due to gravitational waves has been accounted for. This is the quantity, as a multiple of solar masses, which is either within or outside of some sensitivity band for a particular instrument. For example, the chirp masses of Intermediate Mass Black Holes (IMBHs), which is around 10^2-10^5 $M_\odot$, are among the most difficult to detect (3).

The first direct detection of gravitational waves came in 2016 when the Laser Interferometer Gravitational-Wave Observatory (LIGO) confirmed its results were above the 5σ threshold. The GW150194 signal measured came from a black hole of mass $62M_\odot$ whose two components were originally $36M_\odot$ and $29M_\odot$, meaning the energy radiated out in gravitational waves was on the order of $3M_\odot c^2$. (1). Since then, about a dozen more observations have been picked up by LIGO, and with multiple detectors slated for operation in coming decades, this number is expected to grow tremendously.

As such, gravitational wave cosmology is on the precipice of its multi-messenger era, in which it will be possible to explore the nature of transient phenomena, and to construct a statistical portrait of sources such as neutron star mergers, supernovas, and even gamma bursts (4). It is the ongoing goal of gravitational wave cosmology to faithfully map the results of observations and to reconcile these with theory where possible, or else modify theory where necessary.

1.1. *Detectors and telescopes*

The nature of gravitational waves necessitates highly calibrated and sensitive instruments to place reasonable detection limits on signals. LIGO accomplishes this by combining laboratories on different continents into an effectively Earth-sized aperture (1). The LIGO collaboration will soon be joined by Advanced LIGO and KAGRA, which will allow for three-fold higher signal to noise ratio sensitivity (1).

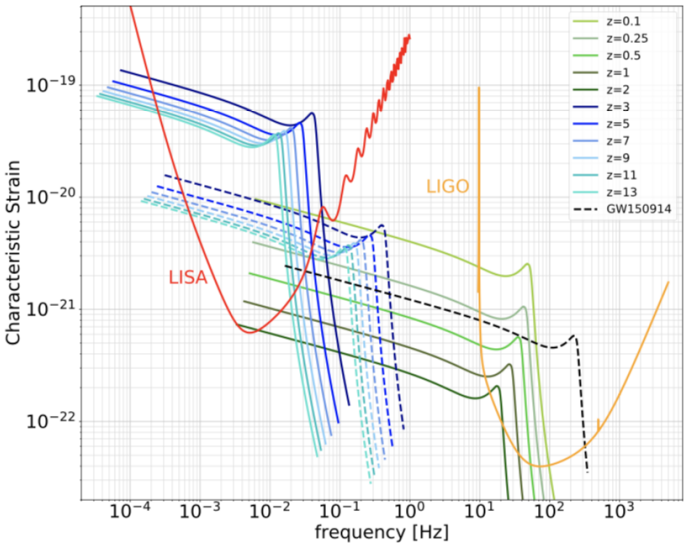

Figure 1 Plot of characteristic strain versus frequency band of IMBHs. The sets of curves describe the inspiral and merger of a given object for a range of masses, redshifts. Notably, the LISA and LIGO bands appear (3)

Pulsar Timing Arrays (PTAs) will study gravitational waves reaching us in the nanohertz frequency band (5)(6) and these, along with Cosmic Microwave Background (CMB) experiments, may finally allow for a signal from the earliest moments after the inflationary phase (7). Approached differently, the absence of such signals once practically possible may lead scientists to dismiss otherwise viable theories (7). Therefore, as it becomes increasingly possible to measure the inflationary-period backgrounds, current theories of the early universe's evolution will be rigorously tested.

The next range in frequencies (shown in Figure 1) to be probed will be the microhertz bands, which will be covered by the Laser Inferometer Space Antenna (LISA), due for launch in 2034 (2). LISA, will look for signals from black holes around the time of the first stars, and directly map the curvature of spacetime around their event horizons. In doing so, it will test the strong lensing interactions (8) around early, quiescent black holes.

Also slated for launch in the 30s are the Einstein Telescope and the Cosmic Explorer, which will look for objects from around the same time as LISA but will be ground-based observatories (2). They will also monitor exotic, compact objects which do not immediately inspiral (9).

2. State of the science

2.1. *Direct evidence for IMBHs*

The first detection of an IMBH earlier this month provided the only existing direct evidence for the formation of black holes in the 100-$10^5 M_\odot$ range (10). In particular, the GW190521 signal detected by the LVK Collaboration is consistent with one of a binary merger of $150 M_\odot$ (10). Long out of reach of traditional detection methods such as accretion disk analysis, IMBHs have proven good candidates for gravitational wave cosmology. Though it will be necessary to increase detection sensitivities to make a statistical portrait possible, LISA will have the capabilities necessary for high z detection on the order of 10^4-10^5 $M_\odot$ mergers (3). IMBHs will remain difficult to detect, but the effort is likely to be worthwhile, as these objects can potentially provide answers to the seeding of supermassive black holes (10), which lie at the centers of galaxies and which help determine their mass and halo distributions.

2.2. *Probing metallicity and heavy element sources*

Gravitational waves also reveal information about the galaxies in which the source objects are contained. For example, it is possible to estimate how likely a detection is in a galaxy given the metallicity compositions of star-forming regions there (11). Low-metallicity environments have a higher incidence of double compact object binary formations resulting in at least one black hole, and gravitational waves are 20 times more likely to be detected from regions with a mix of solar metallicity and heavier element contributions (11). In particular, low-metallicity Population III stars are possible sources of IMBHs, which are predicted to

be the most energetic source of gravitational waves measured by ground-based telescopes detectors (10). These in turn can inform our understanding of the initial mass fractions leading to formation of first seed black holes and reveal whether those are produced by the collapse of early, metal-free stars, or by different processes altogether. Additionally, emissions from neutron star mergers in the infrared have supported the theory claiming heavy elements are produced also by neutron star mergers, and not exclusively by supernovas (12). The detection of gravitational waves from these and other source populations will provide direct evidence for the composition and evolution of these early stars and galaxies.

2.3. *Theories of gravity and nonstandard polarization modes*

Recent direct detection of gravitational waves has also led to exciting deviations of field theories from general relativity at high energies (13)(14) and the possibility of new, nonstandard polarization modes (15).

For example, a cutoff at high energies (within the range observed for GW170817 (16)) necessitates modifications, such as modified gravity, to preserve Lorentz invariance in effective field theories (14). The GW170817 signal was consistent with theory for a large range of frequencies, but not in the band representing the LIGO range, where first order corrections would be necessary to raise the speed of propagation to match that of light (14).

Separately, gravitational waves can also introduce new fundamental physics in describing polarization modes. Though general relativity predicts two types of wave polarization, h_x and h_+, a strong stochastic cosmological gravitational wave background could be expected to contain up to 6 polarization modes (15). Because of its configuration for both frequency and angular modifications, and as a result of its multiple orbiter design, LISA could measure such a background (8)(15). It may also detect other intersting stochastic signals, such as those originating from cosmic strings or early universe phase transitions (17).

3. Project goals

In this project, we hope to explore how the observation in multiple frequency bands of gravitational waves emitted by a set of mid-sized objects can inform our understanding of what happens to these waves between measurements and introduce possible modifications to dispersion relations. By performing a spectral analysis on this class of objects, and studying the frequency-dependent propagation effects, we hope to generate constraints on dispersion relations, explain differences in expected and observed measurements, and develop a phenomenological model.

4. Conclusion

The direct detection in 2015 of gravitational waves from a binary black hole merger (1) a century after Albert Einstein first predicted their existence provided not only a further confirmation of his theory of general relativity, but also the opportunity to further our understanding of fundamental physics, such as early universe phase transitions, evolutionary dynamics, and the structure of gravity itself. As we enter the statistical era of gravitational wave research, exciting new instruments will allow for the probing of exotic objects and developments in theory will continue to inform our understanding of the universe (9)(17). If the first gravitational wave observations have served to confirm theory, then the next set will catalyze it, and in doing so refine our understanding of the universe.

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